

MH2500 PROBABILITY AND INTRODUCTION TO STATISTICS
NANYANG TECHNOLOGICAL UNIVERSITY

Tutorial 9

Tutorial Time: 28/10/2021

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We will discuss several questions from Chapter 7 in Ross' book.

Tutorial Questions

- (1) Consider n independent flips of a coin having probability p of landing on heads. Say that a changeover occurs whenever an outcome differs from the one preceding it. For instance, if $n = 5$ and the outcome is $HHTHT$, then there are 3 changeovers. Find the expected number of changeovers.

Solution: Let $X_i = 1$ if a changeover occurs at position i and $X_i = 0$ otherwise. Then $X = X_2 + X_3 + \cdots + X_n$, and for each i , $P\{X_i = 1\} = p(1-p) + (1-p)p = 2p(1-p)$, as the changeover is either a HT or a TH . As X_i is an indicator variable, $\mathbb{E}(X_i) = P\{X_i = 1\} = 2p(1-p)$. Thus

$$\mathbb{E}(X) = \mathbb{E}(X_2 + \cdots + X_n) = \mathbb{E}(X_2) + \cdots + \mathbb{E}(X_n) = 2(n-1)p(1-p).$$

- (2) The joint density of X and Y is given by

$$f(x, y) = \frac{e^{-y}}{y}, \quad 0 < x < y, 0 < y < \infty$$

Compute $\mathbb{E}[X^3|Y = y]$.

Solution:

$$\mathbb{E}[X^3|Y = y] = \frac{\int_0^y x^3 e^{-y}/y dx}{\int_0^y e^{-y}/y dx} = \frac{(y^4/4) \cdot e^{-y}/y}{e^{-y}} = \frac{1}{4}y^3.$$

- (3) A certain region is inhabited by r distinct types of a certain species of insect. Each insect caught will, independently of the types of the previous catches, be of type i with probability

$$P_i, \quad i = 1, \dots, r, \quad \sum_{i=1}^r P_i = 1.$$

- (a) Compute the mean number of insects that are caught before the first type 1 catch.
(b) Compute the mean number of types of insects that are caught before the first type 1 catch.

Solution:

- (a) Let N be the number of insects that are caught before the first type 1 catch, then $N+1$ is a geometric random variable with parameter P_1 and thus $\mathbb{E}(N+1) = 1/P_1$ and $\mathbb{E}(N) = 1/P_1 - 1$.

(b) Let X_i be the indicator variable that type i is seen before the first type 1 catch, then $X = \sum_{i=2}^r X_i$ is the number of types of insects seen before the first type 1 catch.

We have

$$\begin{aligned}\mathbb{E}(X_i) &= P\{X_i = 1\} = 1 - P\{X_i = 0\} \\ &= 1 - \sum_{n=0}^{\infty} (1 - P_1 - P_i)^n P_1 = 1 - \frac{P_1}{P_1 + P_i} = \frac{P_i}{P_1 + P_i}.\end{aligned}$$

Hence

$$\mathbb{E}(X) = \mathbb{E}\left(\sum_{i=2}^r X_i\right) = \sum_{i=2}^r \mathbb{E}(X_i) = \sum_{i=2}^r \frac{P_i}{P_1 + P_i}.$$

(4) The random variables X and Y have a joint density function given by

$$f(x, y) = \begin{cases} 2e^{-2x}/x, & 0 \leq x < \infty, 0 \leq y \leq x \\ 0, & \text{otherwise} \end{cases}$$

Compute $Cov(X, Y)$.

Solution: Recall $Cov(X, Y) = \mathbb{E}(XY) - \mathbb{E}(X)\mathbb{E}(Y)$. We compute $\mathbb{E}(X), \mathbb{E}(Y), \mathbb{E}(XY)$ as follows:

$$\mathbb{E}(X) = \int_0^{\infty} \int_0^x x \cdot \frac{2e^{-2x}}{x} dy dx = \int_0^{\infty} 2xe^{-2x} dx = \frac{1}{2}.$$

and

$$\mathbb{E}(Y) = \int_0^{\infty} \int_0^x y \cdot \frac{2e^{-2x}}{x} dy dx = \int_0^{\infty} xe^{-2x} dx = \frac{1}{4}.$$

Also

$$\mathbb{E}(XY) = \int_0^{\infty} \int_0^x xy \cdot \frac{2e^{-2x}}{x} dy dx = \int_0^{\infty} x^2 e^{-2x} dx = \frac{1}{4},$$

Thus,

$$Cov(X, Y) = \mathbb{E}(XY) - \mathbb{E}(X)\mathbb{E}(Y) = \frac{1}{4} - \left(\frac{1}{2}\right)\left(\frac{1}{4}\right) = \frac{1}{8}.$$

(5) The joint density function of X and Y is given by

$$f(x, y) = \frac{1}{y} e^{-(y+x/y)}, \quad x > 0, y > 0$$

Find $\mathbb{E}(X)$, $\mathbb{E}(Y)$, and show that $Cov(X, Y) = 1$.

Solution: We can compute that

$$\begin{aligned}\mathbb{E}(X) &= \int_0^{\infty} \int_0^{\infty} x \cdot \frac{1}{y} e^{-(y+x/y)} dy dx = \int_0^{\infty} \frac{e^{-y}}{y} \int_0^{\infty} xe^{-x/y} dx dy \\ &= \int_0^{\infty} \frac{e^{-y}}{y} y^2 dy = 1,\end{aligned}$$

$$\begin{aligned}\mathbb{E}(Y) &= \int_0^{\infty} \int_0^{\infty} y \cdot \frac{1}{y} e^{-(y+x/y)} dy dx = \int_0^{\infty} e^{-y} \int_0^{\infty} e^{-x/y} dx dy \\ &= \int_0^{\infty} e^{-y} y dy = 1,\end{aligned}$$

and

$$\begin{aligned}\mathbb{E}(XY) &= \int_0^\infty \int_0^\infty xy \cdot \frac{1}{y} e^{-(y+x/y)} dy dx = \int_0^\infty e^{-y} \int_0^\infty x e^{-x/y} dx dy \\ &= \int_0^\infty e^{-y} y^2 dy \\ &= 2.\end{aligned}$$

Therefore

$$Cov(X, Y) = \mathbb{E}(XY) - \mathbb{E}(X) \mathbb{E}(Y) = 2 - 1 \times 1 = 1.$$